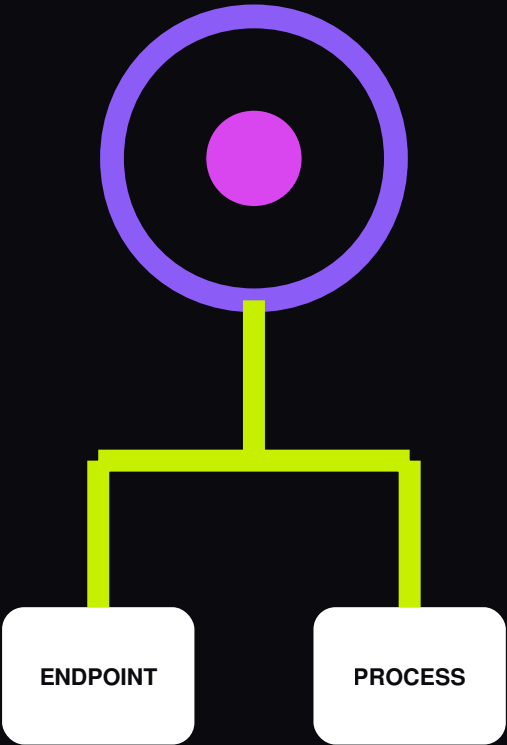


# SSH LOCAL COMMAND VIA CMD.EXE

Hunts for ssh.exe launched with both PermitLocalCommand=yes and LocalCommand=cmd.exe, an option pairing observed in RedFlick delivery.

HUNTING QUERY

CONFIDENCE: MEDIUM



EVIDENCE BEFORE ALERTING

|                        |                                |
|------------------------|--------------------------------|
| REPORT DATE            | STATUS                         |
| 30 September 2026      | Untested implementation sketch |
| PRIMARY PLATFORM       | OPERATIONAL VALUE              |
| Microsoft Defender XDR | 9.6 / 10                       |



# EXECUTIVE ASSESSMENT

Decision, confidence and evidence boundary.

ASSESSMENT

## OPERATIONAL VALUE

9.6

Hunts for ssh.exe launched with both PermitLocalCommand=yes and LocalCommand=cmd.exe, an option pairing observed in RedFlick delivery.

## STATUS

### Hunting query

Untested implementation sketch; Confidence Medium.

1primary table

0joins

0secondary rules

2validation gates

### Evidence boundary

A result proves only that ssh.exe received the two local-command options. It does not prove Star Blizzard, a successful connection, MSI execution, CosmicPulse installation, or malicious intent.



# BEHAVIOR AND EVIDENCE

Observed facts separated from analytical scope.

MECHANICS

## # Source-supported fact

- 1 Microsoft observed Star Blizzard RedFlick campaigns from January through August 2026 and published the analysis on September 29.
- 2 In the January VHDX chain, a BAT file invoked ssh.exe with PermitLocalCommand enabled to download and execute a remote MSI.
- 3 Microsoft published an exact Defender XDR hunt for PermitLocalCommand=yes together with LocalCommand=cmd.exe and warned that matches may be benign.

### Detection hypothesis

If an attacker repeats this behavior, endpoint process telemetry should expose the same source-supported process or command-line pattern. Distinction requires ownership, asset, command, and approved-change context.

### Unconfirmed elements

A result proves only that ssh.exe received the two local-command options. It does not prove Star Blizzard, a successful connection, MSI execution, CosmicPulse installation, or malicious intent.

**Logic:** Behavioral command-line pattern **ATT&CK:** T1059.003 / T1021.004



# TELEMETRY CONTRACT

Collection prerequisites and field reliability.

TELEMETRY

REQUIRED

Microsoft Defender for Endpoint process creation telemetry with complete ssh.exe command line on Windows endpoints.

OPTIONAL ENRICHMENT

SSH destination, source archive/LNK/VHDX evidence, file-origin metadata, MSI creation/execution, email attachment lineage, and change records.

| Fields                                    | Reliability           | Use             |
|-------------------------------------------|-----------------------|-----------------|
| Timestamp, DeviceId, DeviceName           | Documented            | Query or triage |
| FileName, FolderPath, ProcessCommandLine  | Documented            | Query or triage |
| AccountName, ProcessId, ProcessUniqueId   | Documented            | Query or triage |
| InitiatingProcessFileName                 | Documented            | Query or triage |
| InitiatingProcessCommandLine              | Documented            | Query or triage |
| InitiatingProcessParentFileName, ReportId | Documented            | Query or triage |
| Approved SSH automation inventory         | Environment-dependent | Query or triage |

Excluded

No payload content, inferred execution success, actor attribution, IOC dependency, or undocumented cross-SIEM mapping is used.



# PRIMARY KQL

Direct source-supported conditions with investigable output.

HUNTING

STATUS

Hunting query — Untested implementation sketch

KQL — UNTESTED

```
DeviceProcessEvents
| where FileName =~ "ssh.exe"
| where ProcessCommandLine has "PermitLocalCommand=yes"
| where ProcessCommandLine has "LocalCommand=cmd.exe"
| project Timestamp, DeviceId, DeviceName, FileName,
    FolderPath, ProcessCommandLine, AccountName,
    ProcessId, ProcessUniqueId, InitiatingProcessFileName,
    InitiatingProcessCommandLine,
    InitiatingProcessParentFileName, ReportId
| order by Timestamp desc
```

WINDOW

Analyst-selected;  
none embedded

THRESHOLD

None

JOIN / SEQUENCE

None

KEY

Single process  
event



# TRIAGE AND TUNING

Human validation before response.

WORKFLOW

1

## Preserve

Retain the raw process event, paths, command lines, accounts, process identifiers, timestamp, and ReportId.

2

## Confirm

Identify the device owner, executing account, software owner, approved automation, and matching change record.

3

## Enrich

SSH destination, source archive/LNK/VHDX evidence, file-origin metadata, MSI creation/execution, email attachment lineage, and change records.

4

## Scope

Review adjacent process, service, file, network, email-delivery, and software-deployment evidence.

### Expected benign overlap

Developer tooling, SSH configuration tests, and approved automation may use LocalCommand. Review exact option pairing, destination, account, host owner, SSH path, and change evidence; do not tune by rarity alone.

### Response boundary

A result proves only that ssh.exe received the two local-command options. It does not prove Star Blizzard, a successful connection, MSI execution, CosmicPulse installation, or malicious intent.



# VALIDATION GATES

Controlled tests and cross-SIEM limits.

QUALITY CONTROL

## Validation steps

- 1. Run an authorized ssh.exe test with both options and retain the raw Defender process event.
- 2. Confirm the full command line, executable path, account, parent process, and ProcessUniqueld are populated.
- 3. Measure benign overlap by device and account over a representative local window.
- 4. Collect destination and adjacent file/MSI events for triage without making them mandatory detection conditions.

## Decision criteria

**Success:** The authorized test is returned with both exact options and sufficient ownership context to investigate matches.

**Rejection:** The option pairing is common in approved automation and no reliable destination, owner, or command tuning dimension exists.

**Failure:** Command line truncation, options stored only in config, renamed SSH client, non-Windows implementation, or endpoint telemetry gaps.

| Platform     | Decision                   | Missing evidence                  |
|--------------|----------------------------|-----------------------------------|
| Defender XDR | Hunting query              | Endpoint baseline and validation  |
| Splunk       | No reliable implementation | Index, sourcetype, parser, sample |
| Chronicle    | No reliable implementation | Parser and exact UDM mapping      |



# CONCLUSION AND REFERENCES

Coverage statement and document control.

FINAL

## CONCLUSION

### Hunting query

Hunts for ssh.exe launched with both PermitLocalCommand=yes and LocalCommand=cmd.exe, an option pairing observed in RedFlick delivery.

A result proves only that ssh.exe received the two local-command options. It does not prove Star Blizzard, a successful connection, MSI execution, CosmicPulse installation, or malicious intent.

## MITRE ATT&CK

T1059.003 / T1021.004

## References

1. Microsoft Threat Intelligence — RedFlick — 29 Sep 2026
2. Microsoft Learn — DeviceProcessEvents — updated 7 Aug 2026
3. MITRE ATT&CK — T1059.003 and T1021.004

## Document control

English dossier. No customer data. URLs are defanged in references.txt. HTML and rendering assets remain private.